

■ Stochastic Modeling (622) Cheat Sheet

1. Random Variables & Distributions

Bernoulli, Binomial, Poisson, Normal, Exponential
Mean (μ), Variance (σ^2), Standard Deviation (σ)
PDF vs CDF

2. Conditional Probability & Bayes' Theorem

$$P(A|B) = P(A \cap B) / P(B)$$
$$P(A|B) = [P(B|A)P(A)] / P(B)$$

3. Expectation & Variance

$$E[X] = \sum x P(X=x) \text{ (discrete)}, E[X] = \int x f(x) dx \text{ (continuous)}$$
$$\text{Var}(X) = E[X^2] - (E[X])^2$$

4. Covariance & Correlation

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$
$$\rho = \text{Cov}(X,Y) / (\sigma_X \sigma_Y)$$

5. Law of Large Numbers & Central Limit Theorem

LLN: Sample average $\rightarrow$ expected value as $n \rightarrow \infty$
CLT: $\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n

6. Markov Chains

Transition matrix $P = [p_{ij}]$
Chapman–Kolmogorov: $P^{(n+m)} = P^n P^m$
Stationary distribution π : $\pi P = \pi$

7. Poisson Process

$N(t)$: number of arrivals by time t
Stationary & independent increments
 $P(N(t)=k) = (\lambda t)^k e^{-(\lambda t)} / k!$

8. Brownian Motion (Wiener Process)

$W(0)=0$, continuous paths
Independent, normally distributed increments
 $W(t) - W(s) \sim N(0, t-s)$

9. Ito's Lemma (SDEs)

If $dX = \mu(X,t)dt + \sigma(X,t)dW$, $Y=f(X,t)$:

$$dY = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW$$

10. Applications in Finance

Geometric Brownian Motion: $dS = \mu S dt + \sigma S dW$

Black–Scholes (Call Option): $C = S N(d) - K e^{-rt} N(d)$